In [ ]:

```
#Name: Meenakshi Sangeeth
#Registration No: 240905630
#Section: PG
#Roll No: 34
```

In [2]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [3]:

```
df=pd.read_csv('mtcars.csv')
pd.set_option('display.max_columns', 100)  #display all columns
pd.set_option('display.max_rows',None)      #display all rows
```

In [4]:

```
df
```

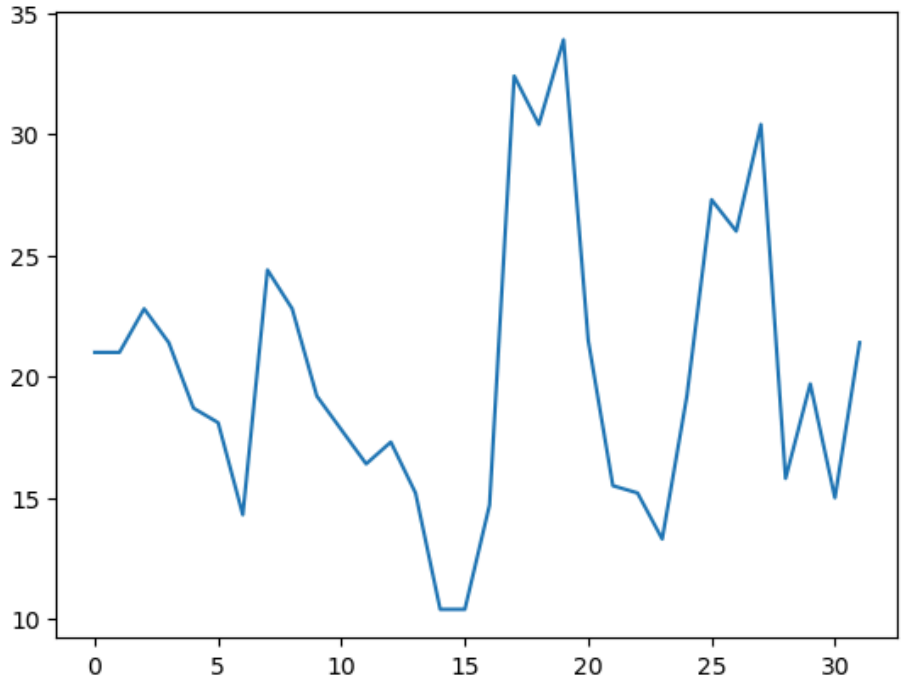
Out[4]:

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
0	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
2	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
3	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
4	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
5	Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
6	Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
7	Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
8	Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
9	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
10	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
11	Merc 450SE	16.4	8	275.8	180	3.07	4.070	17.40	0	0	3	3
12	Merc 450SL	17.3	8	275.8	180	3.07	3.730	17.60	0	0	3	3
13	Merc 450SLC	15.2	8	275.8	180	3.07	3.780	18.00	0	0	3	3
14	Cadillac Fleetwood	10.4	8	472.0	205	2.93	5.250	17.98	0	0	3	4
15	Lincoln Continental	10.4	8	460.0	215	3.00	5.424	17.82	0	0	3	4
16	Chrysler Imperial	14.7	8	440.0	230	3.23	5.345	17.42	0	0	3	4
17	Fiat 128	32.4	4	78.7	66	4.08	2.200	19.47	1	1	4	1
18	Honda Civic	30.4	4	75.7	52	4.93	1.615	18.52	1	1	4	2
19	Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.90	1	1	4	1
20	Toyota Corona	21.5	4	120.1	97	3.70	2.465	20.01	1	0	3	1
21	Dodge Challenger	15.5	8	318.0	150	2.76	3.520	16.87	0	0	3	2
22	AMC Javelin	15.2	8	304.0	150	3.15	3.435	17.30	0	0	3	2
23	Camaro Z28	13.3	8	350.0	245	3.73	3.840	15.41	0	0	3	4
24	Pontiac Firebird	19.2	8	400.0	175	3.08	3.845	17.05	0	0	3	2
25	Fiat X1-9	27.3	4	79.0	66	4.08	1.935	18.90	1	1	4	1
26	Porsche 914-2	26.0	4	120.3	91	4.43	2.140	16.70	0	1	5	2
27	Lotus Europa	30.4	4	95.1	113	3.77	1.513	16.90	1	1	5	2
28	Ford Pantera L	15.8	8	351.0	264	4.22	3.170	14.50	0	1	5	4
29	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6
30	Maserati Bora	15.0	8	301.0	335	3.54	3.570	14.60	0	1	5	8
31	Volvo 142E	21.4	4	121.0	109	4.11	2.780	18.60	1	1	4	2

In [5]:

```
# 1. Extract the 'mpg' column from the dataframe & Plot the mpg values as a line plot
mpg = df['mpg']
mpg.plot()
```

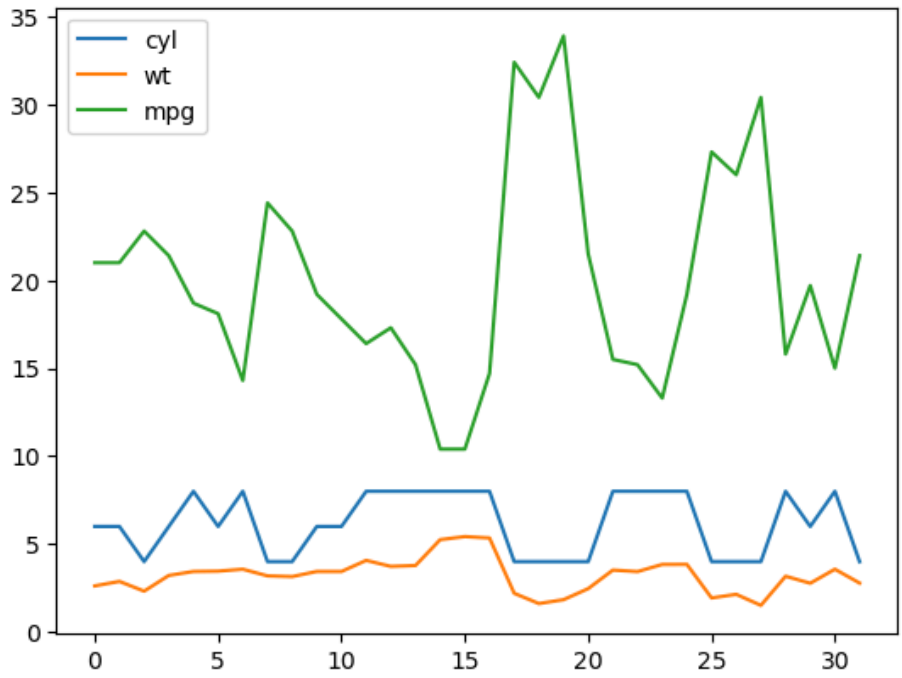
Out[5]: <AxesSubplot:>



In [6]:

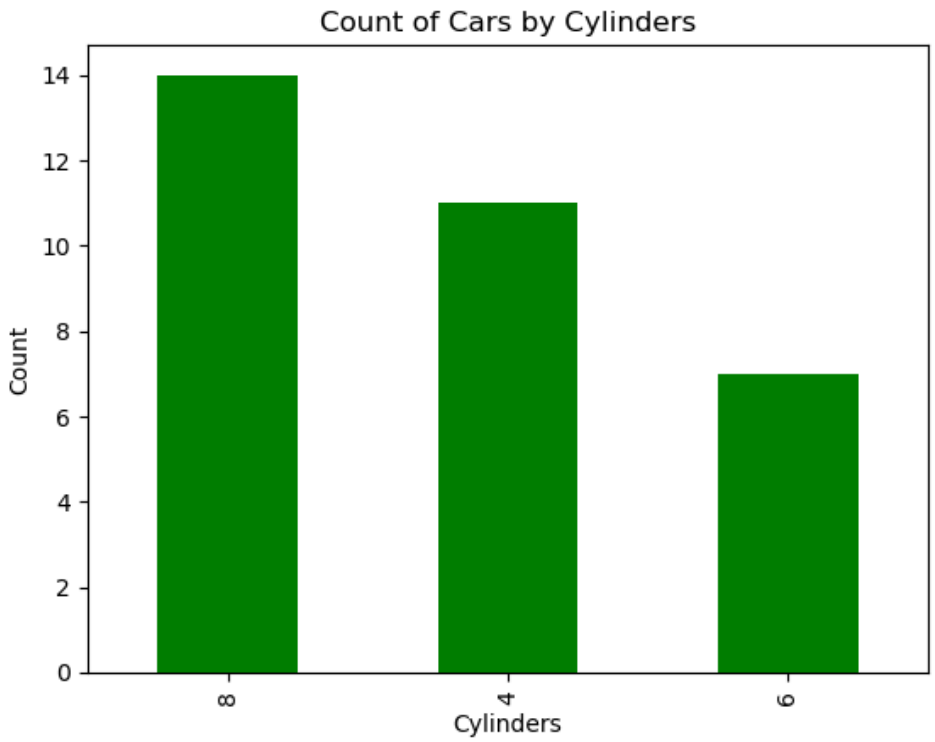
```
# 2. Select only the 'cyl', 'wt', and 'mpg' columns from the dataframe & Plot the selected
df = df[['cyl', 'wt', 'mpg']]
df.plot()
```

Out[6]: <AxesSubplot:>



In [7]:

```
#3. Bar Plot- the count of cars by the number of cylinders
cyl_counts = df['cyl'].value_counts()
cyl_counts.plot(kind='bar', color='green')
plt.title('Count of Cars by Cylinders')
plt.xlabel('Cylinders')
plt.ylabel('Count')
plt.show()
```

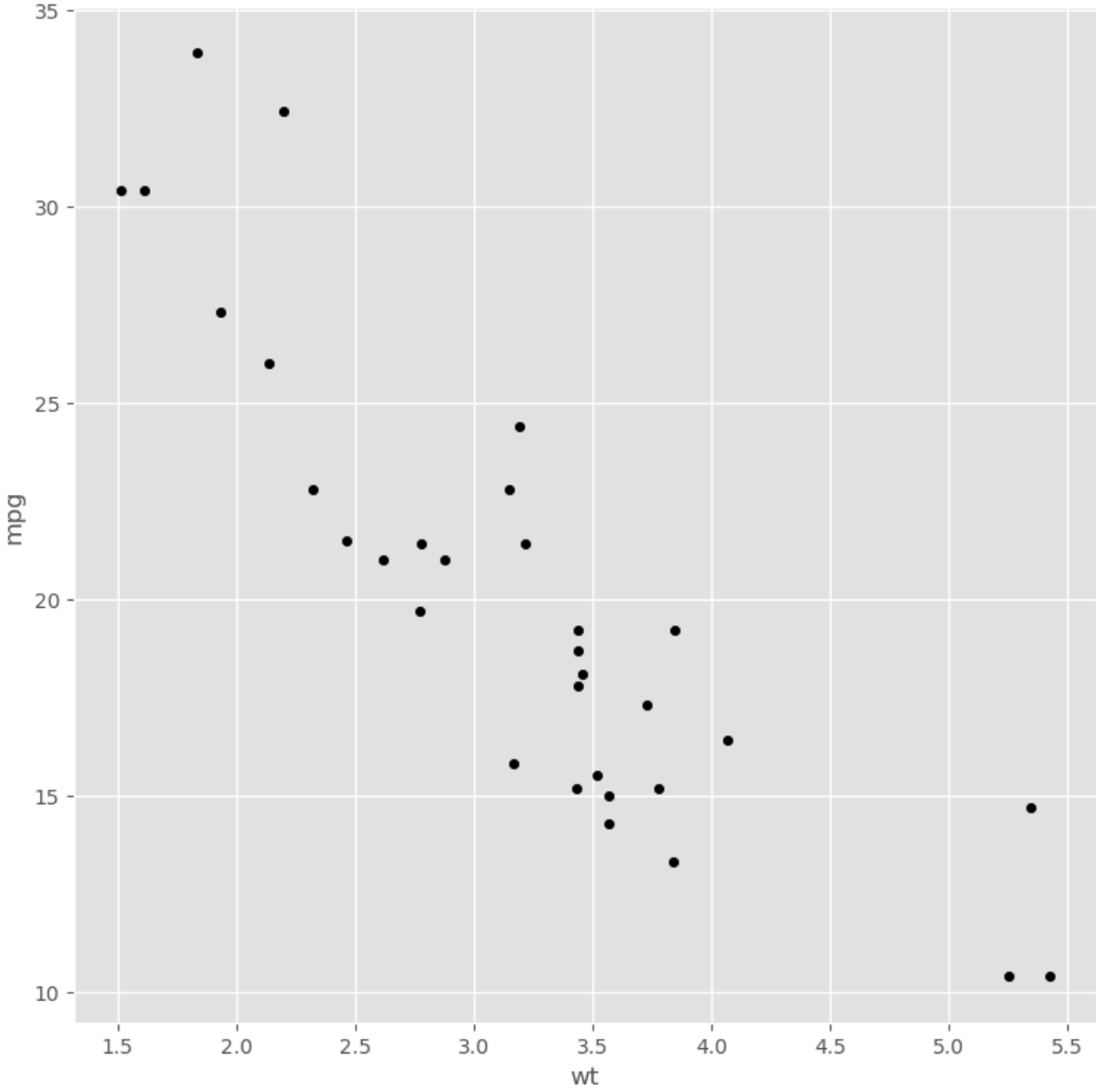


In [8]:

```
plt.style.use('ggplot')
```

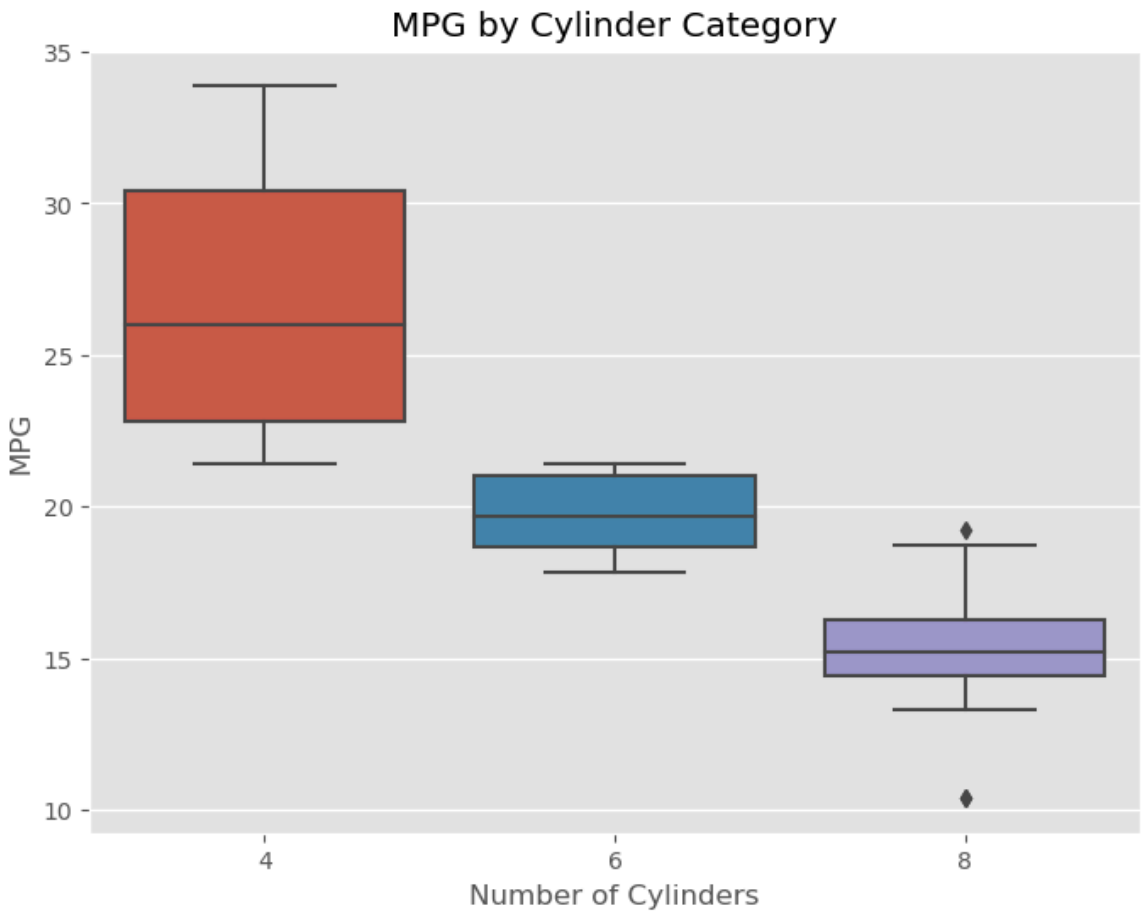
In [9]:

```
#4. Scatter plot of weight vs mpg
df.plot(
    kind="scatter",
    x="wt",
    y="mpg",
    figsize=(9,9),
    color="black",
);
```



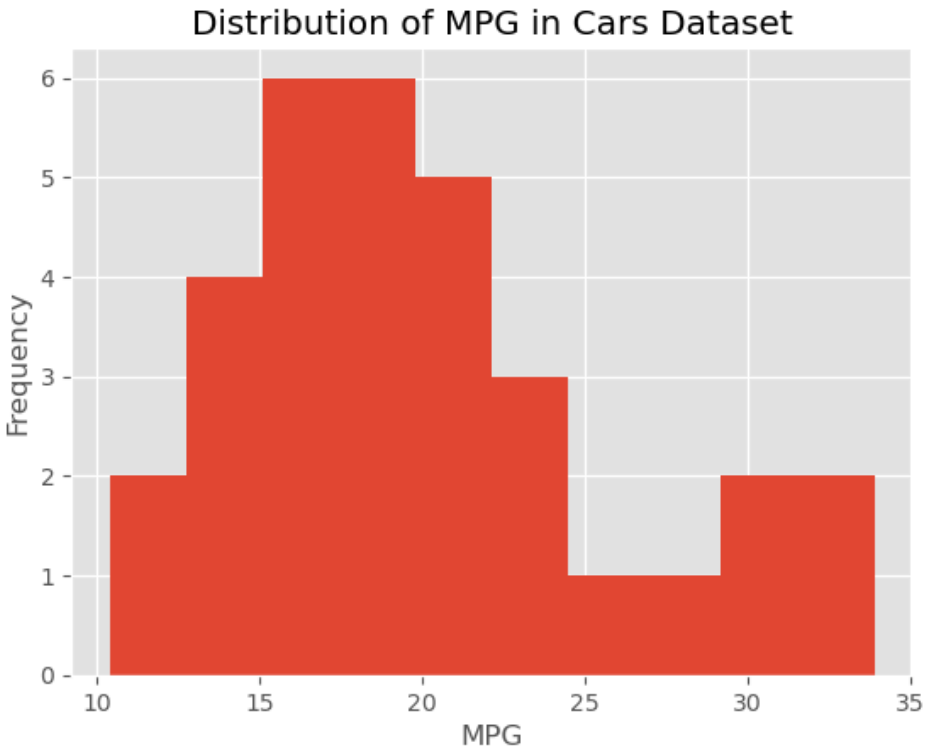
In [10]:

```
#5. Box plot of MPG for each cylinder category
plt.figure(figsize=(8, 6))
sns.boxplot(x='cyl', y='mpg', data=df)
plt.title('MPG by Cylinder Category')
plt.xlabel('Number of Cylinders')
plt.ylabel('MPG')
plt.show()
```



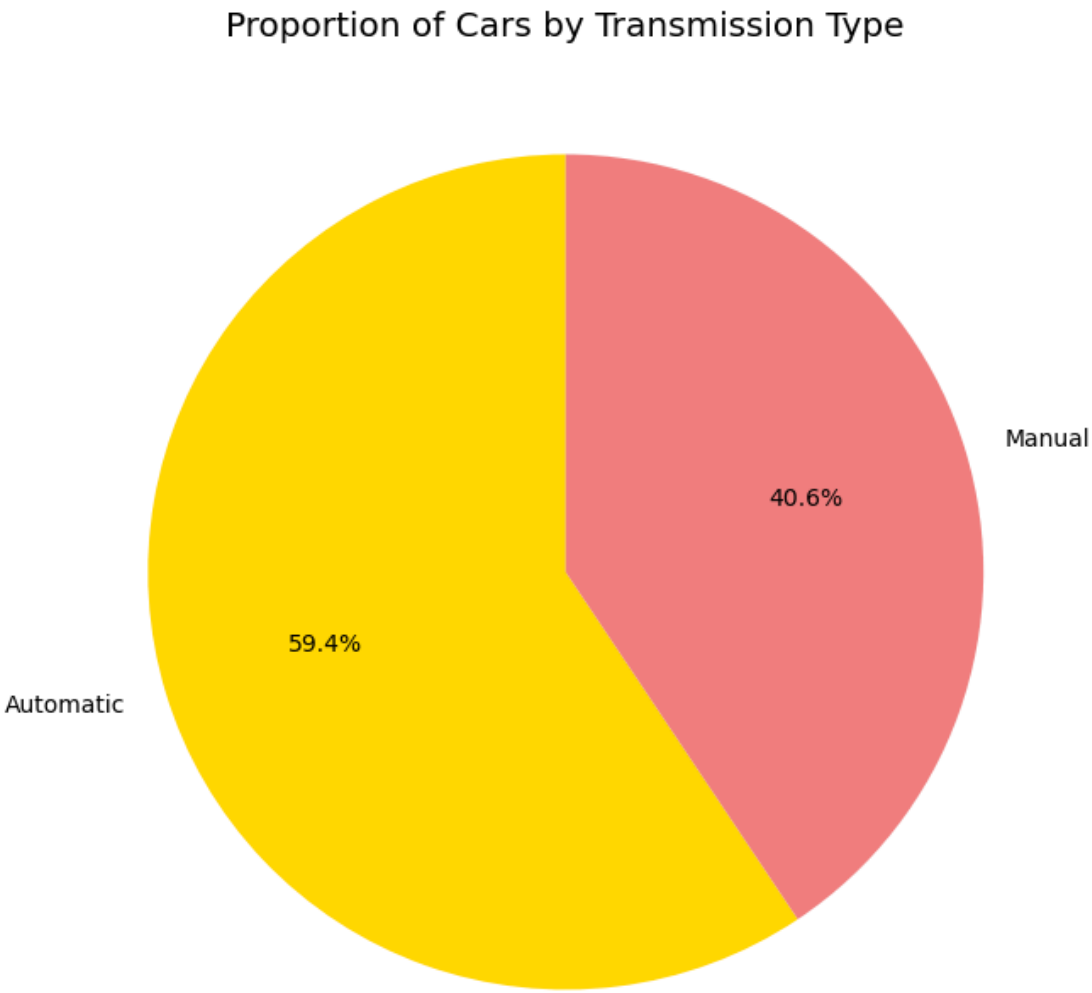
In [11]:

```python
# 6. Histogram of MPG for all cars

x = df['mpg']

# Plotting the histogram
plt.hist(x)

# Adding title to the plot
plt.title("Distribution of MPG in Cars Dataset")

# Adding label on the y-axis (Frequency of each cylinder count)
plt.ylabel('Frequency')

# Adding label on the x-axis (Number of cylinders in cars)
plt.xlabel('MPG')

# Displaying the plot
plt.show()
```



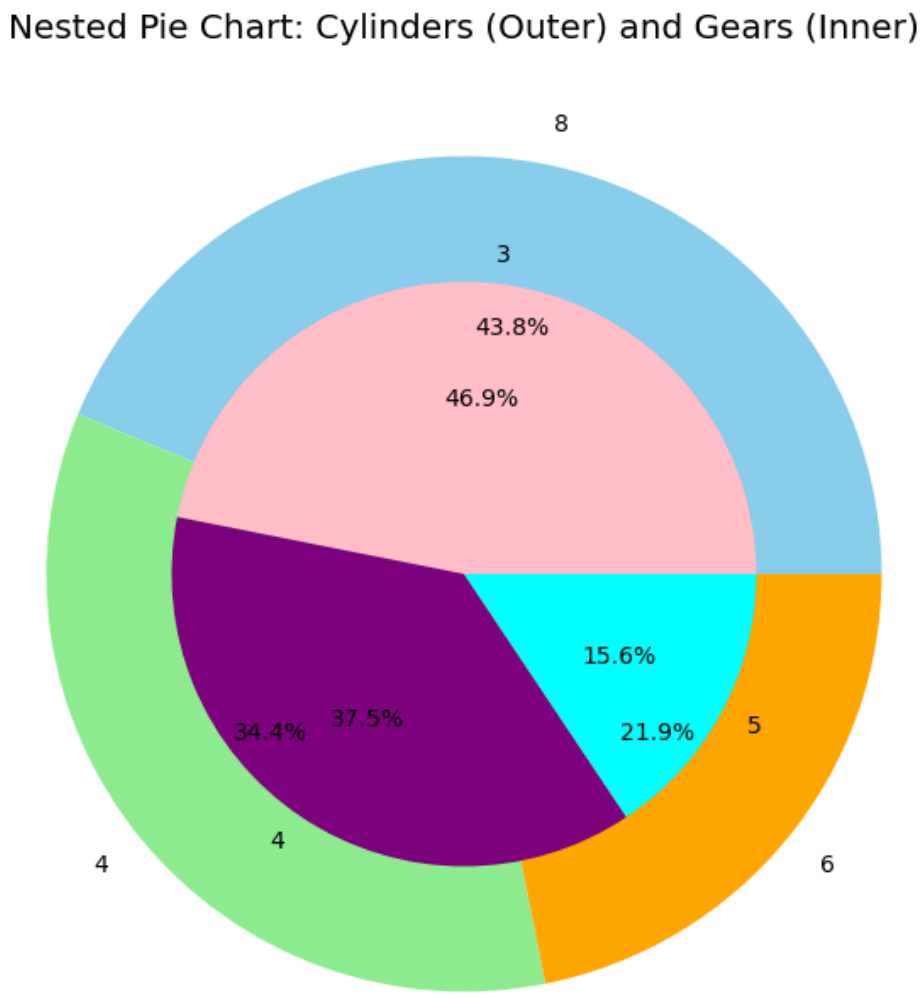
In [15]:

```python
#7. Pie Chart for Transmission Types

df = pd.read_csv('mtcars.csv')


am_counts = df['am'].value_counts()

plt.figure(figsize=(8, 8))
plt.pie(
    am_counts,
    labels=['Automatic', 'Manual'],
    autopct='%1.1f%%',
    colors=['gold', 'lightcoral'],
    startangle=90
)
plt.title('Proportion of Cars by Transmission Type')
plt.show()
```



In [16]:

```
# 8. Nested Pie Chart which needs Data preparation
outer_labels = df['cyl'].value_counts().index
outer_sizes = df['cyl'].value_counts().values

inner_labels = df['gear'].value_counts().index
inner_sizes = df['gear'].value_counts().values

fig, ax = plt.subplots(figsize=(8, 8))
ax.pie(outer_sizes, labels=outer_labels, radius=1, autopct='%1.1f%%', colors=['skyblue', '
ax.pie(inner_sizes, labels=inner_labels, radius=0.7, autopct='%1.1f%%', colors=['pink', 'p
ax.set(aspect="equal")
plt.title('Nested Pie Chart: Cylinders (Outer) and Gears (Inner)')
plt.show()
```



In [ ]:

```
#Exercise 2
```

In [41]:

```
df = pd.read_csv("matches.csv")
df
```

15	16	2017	Mumbai	2017-04-16	Gujarat Lions	Mumbai Indians	Mumbai Indians	field	normal	0	
16	17	2017	Bangalore	2017-04-16	Rising Pune Supergiant	Royal Challengers Bangalore	Royal Challengers Bangalore	field	normal	0	
17	18	2017	Delhi	2017-04-17	Delhi Daredevils	Kolkata Knight Riders	Delhi Daredevils	bat	normal	0	
18	19	2017	Hyderabad	2017-04-17	Sunrisers Hyderabad	Kings XI Punjab	Kings XI Punjab	field	normal	0	
19	20	2017	Rajkot	2017-04-18	Royal Challengers Bangalore	Gujarat Lions	Gujarat Lions	field	normal	0	C

In [18]:

```
#1.Find the names of teams that are part of IPL
teams = set(df['team1']).union(set(df['team2']))
teams = list(teams)
teams
```

Out[18]:

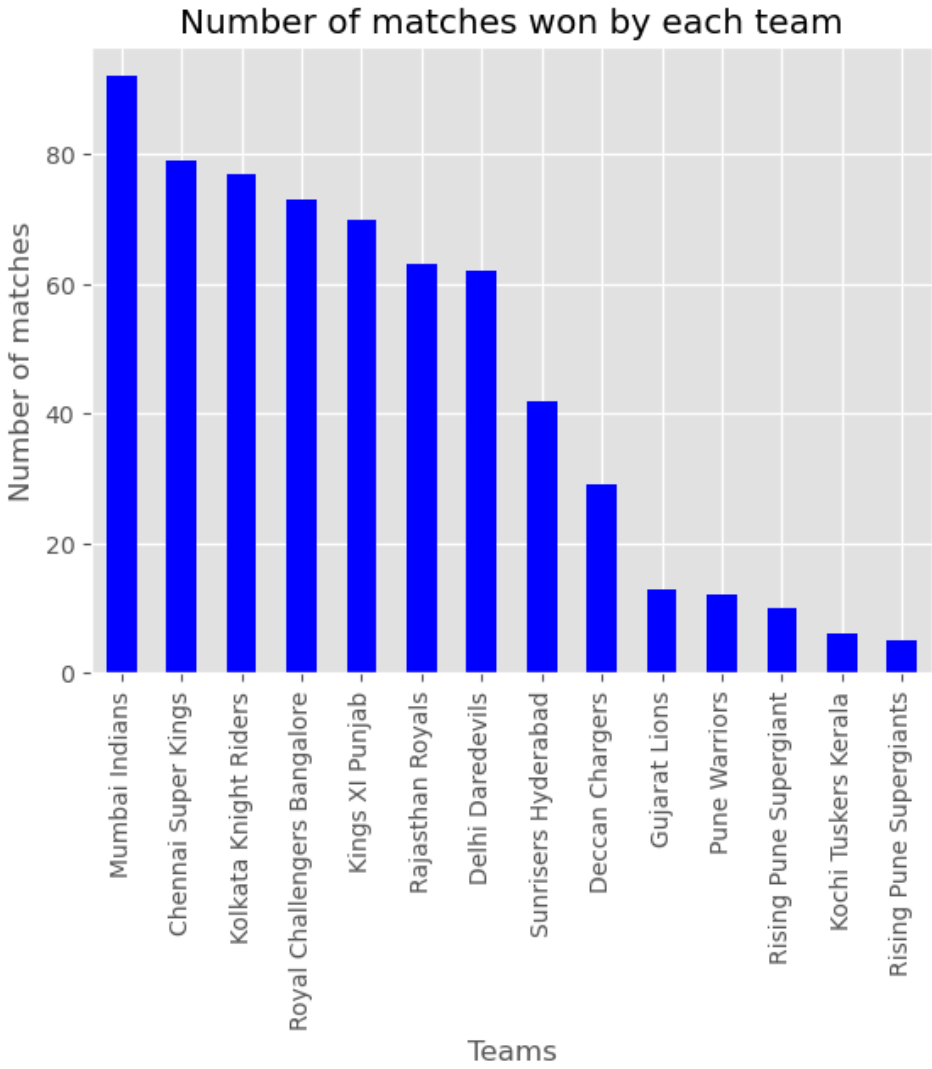
```
['Deccan Chargers',
 'Kochi Tuskers Kerala',
 'Delhi Daredevils',
 'Rajasthan Royals',
 'Gujarat Lions',
 'Mumbai Indians',
 'Kolkata Knight Riders',
 'Sunrisers Hyderabad',
 'Rising Pune Supergiant',
 'Pune Warriors',
 'Kings XI Punjab',
 'Royal Challengers Bangalore',
 'Chennai Super Kings',
 'Rising Pune Supergiants']
```

In [45]:

```
#2. Visualize the total number of matches won by each team

wins = df["winner"].value_counts()
wins.plot(kind='bar', title = 'Number of matches won by each team', color='blue', xlabel="
          ylabel='Number of matches')
```

Out[45]:

```
<AxesSubplot:title={'center':'Number of matches won by each team'}, xlabel='Teams', ylabel='N
umber of matches'>
```



In [21]:

```
#3. List the names of umpires who have served as first umpire and the number of matches
df["umpire2"].value_counts()
```

Out[21]:

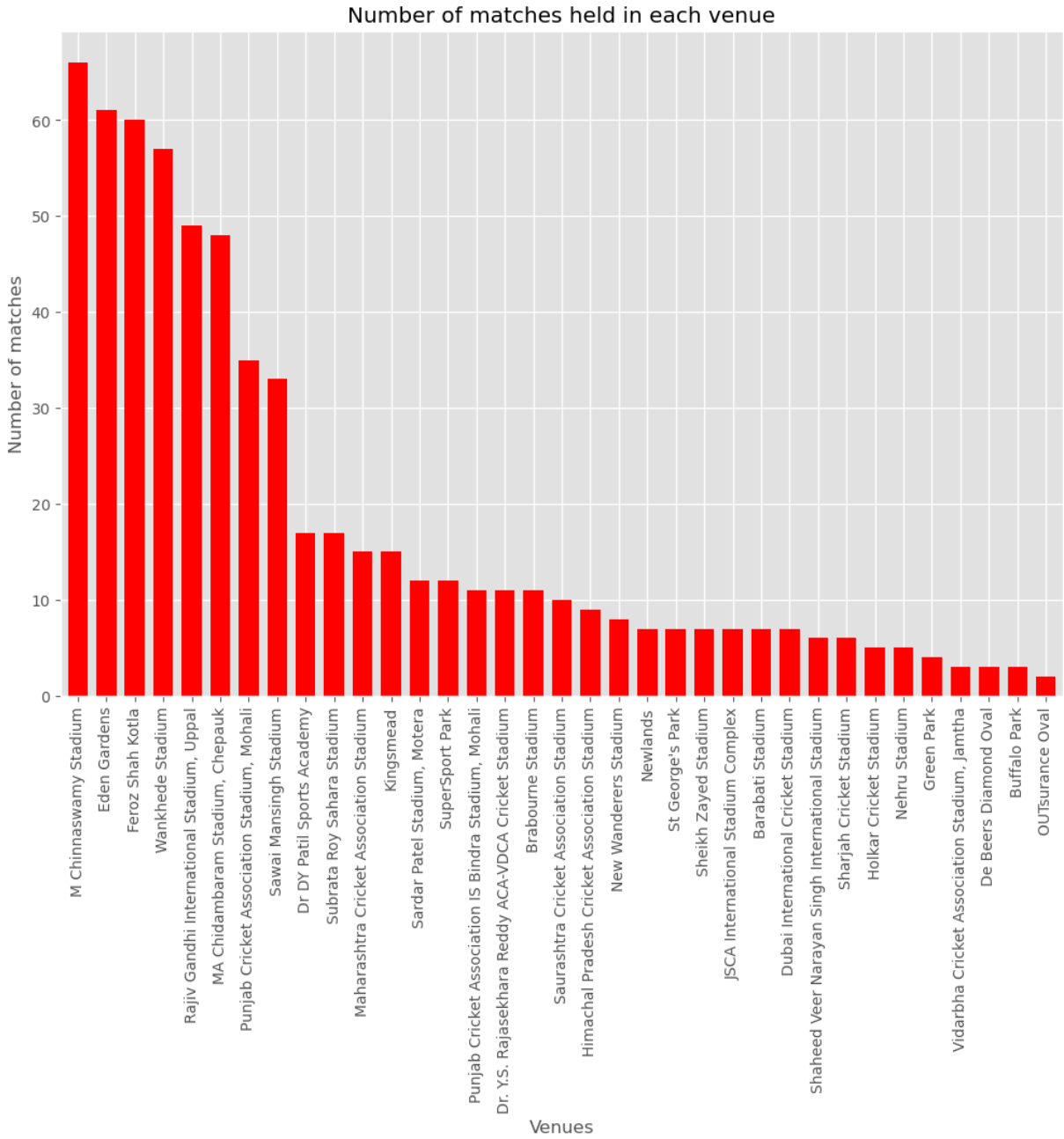
```
SJA Taufel          54
C Shamshuddin       50
S Ravi              49
RJ Tucker           38
CK Nandan           36
RB Tiffin           30
VA Kulkarni         28
SK Tarapore         26
BNJ Oxenford        25
AM Saheba           23
M Erasmus           22
RE Koertzen         21
DJ Harper           18
AK Chaudhary        15
HDPK Dharmasena     14
NJ Llong            13
K Srinath           13
SL Shastri          13
K Hariharan         11
S Asnani            11
CB Gaffaney         10
VK Sharma            9
PR Reiffel           8
RK Illingworth       8
BR Doctrove          8
Nitin Menon          8
IL Howell            7
I Shivram            6
SS Hazare            6
AL Hill              6
GA Pratapkumar       6
PG Pathak            6
TH Wijewardene       5
A Nand Kishore       5
S Das                5
BG Jerling           4
AV Jayaprakash       4
K Srinivasan         3
A Deshmukh           3
SD Ranade            2
MR Benson            2
JD Cloete            1
Subroto Das          1
SJ Davis             1
SD Fry               1
Name: umpire2, dtype: int64
```

In [46]:

```
#4. Visualize how many matches were held in each venue
venue = df["venue"].value_counts()
plt.figure(figsize=(12, 8))
venue.plot(kind='bar', title = 'Number of matches held in each venue', color='red',
          width=.7, xlabel="Venues", ylabel='Number of matches')

```

Out[46]:

```
<AxesSubplot:title={'center':'Number of matches held in each venue'}, xlabel='Venues', ylabel
='Number of matches'>
```

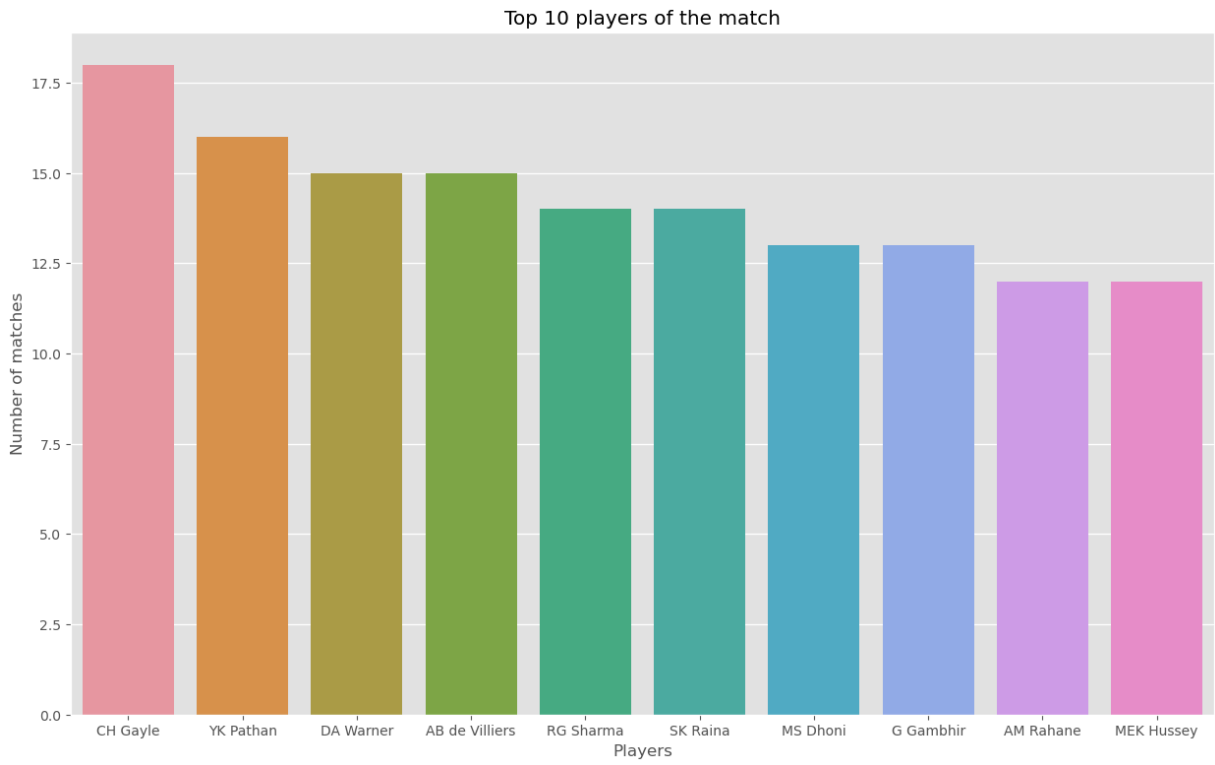


In [48]:

```
#5. Visualize the top 10 players identified as the player of the match for the most signif
top = df["player_of_match"].value_counts().head(10)
plt.figure(figsize=(15,9))
sns.barplot(x=top.index, y=top.values)
plt.title('Top 10 players of the match')
plt.xlabel('Players')
plt.ylabel('Number of matches')
plt
```

Out[48]:

```
<module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-packages\\matplotlib
\\pyplot.py'>
```

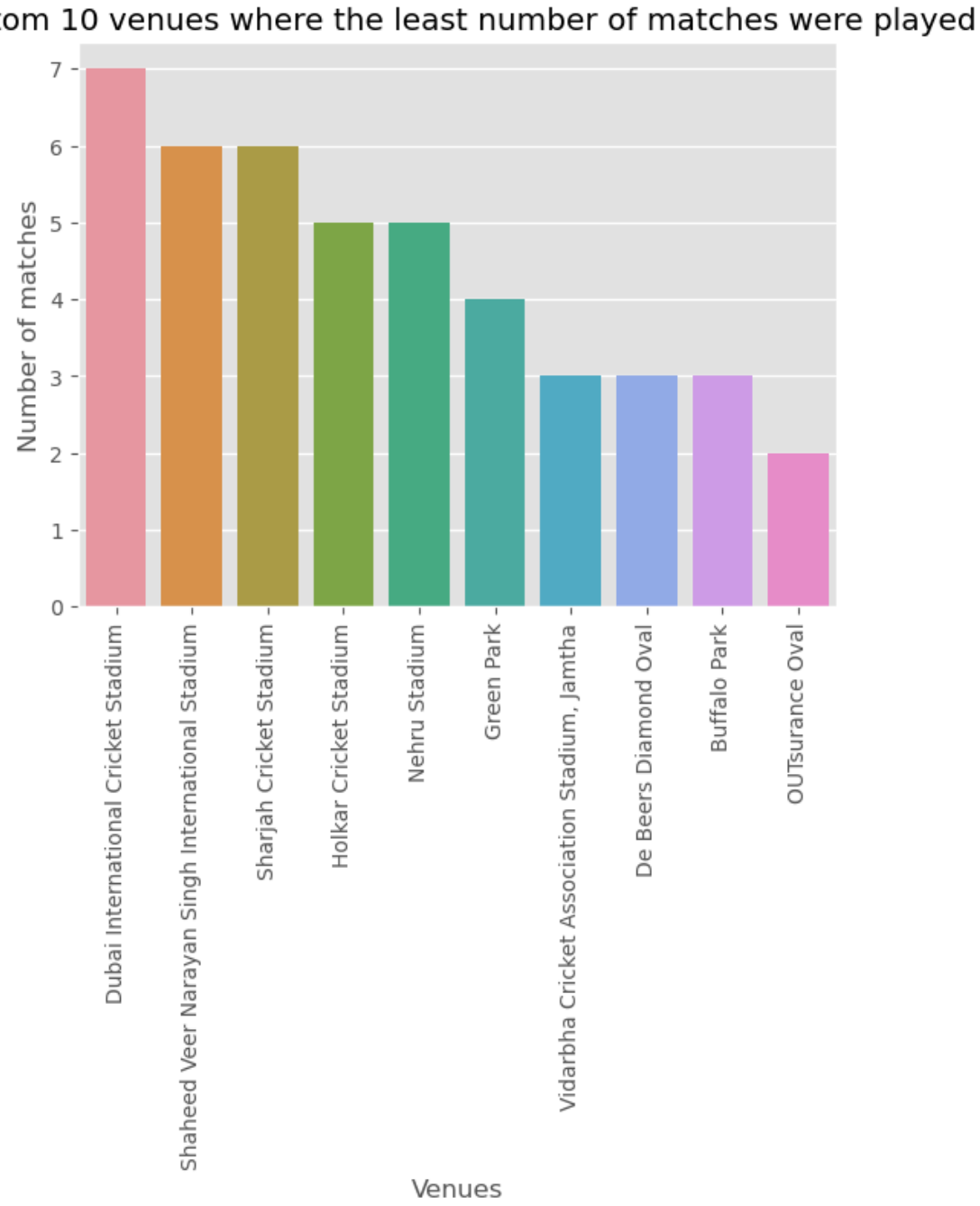


In [52]:

```
#6. Visualize the bottom 10 venues where the least number of matches were played.
ven = df["venue"].value_counts().tail(10)
sns.barplot(x=ven.index, y=ven.values)
plt.title('Bottom 10 venues where the least number of matches were played')
plt.xlabel('Venues')
plt.ylabel('Number of matches')
plt.xticks(rotation=90)
plt
```

Out[52]:

```
<module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-packages\\matplotlib
\\pyplot.py'>
```

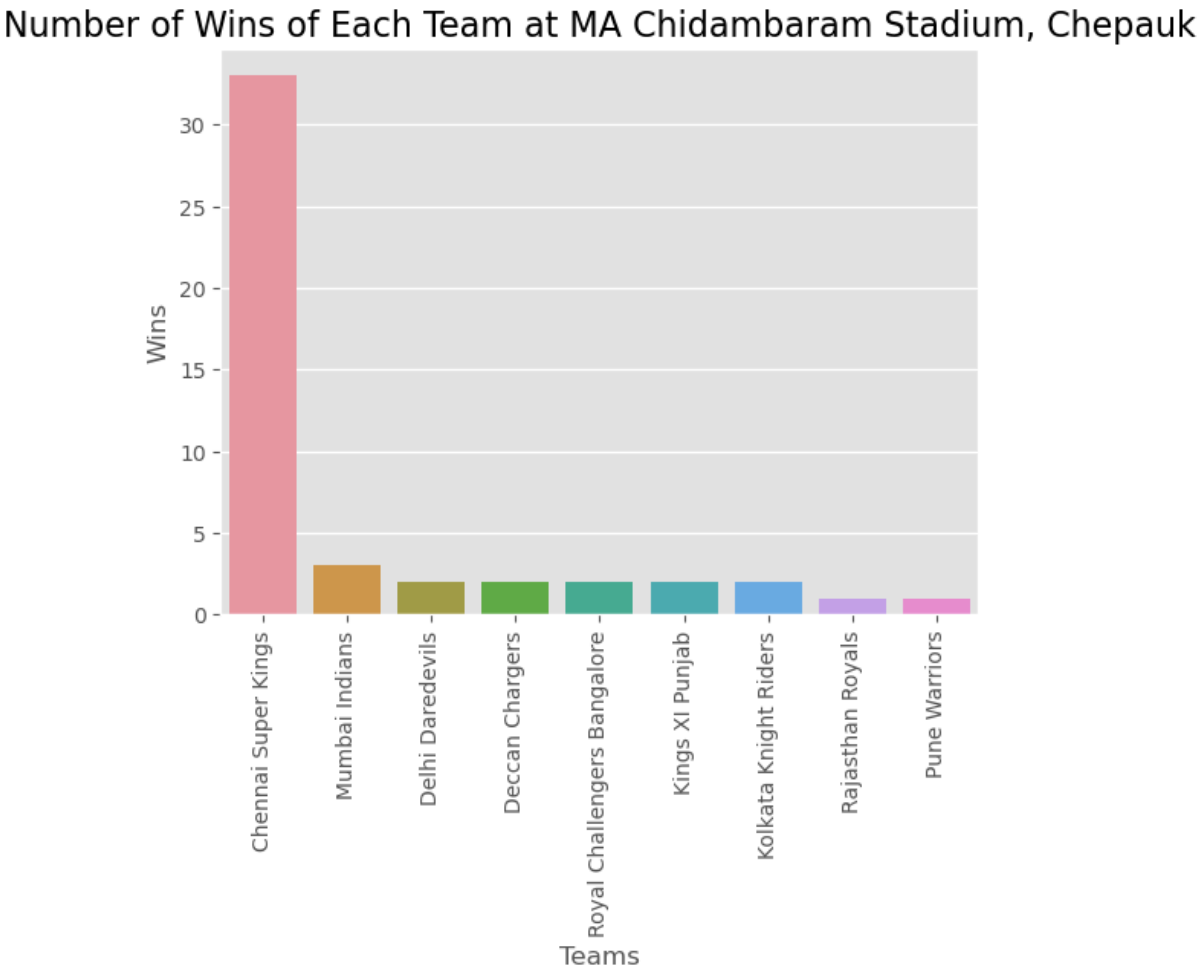


In [54]:

```
#7. Visualize the number of wins of each team for the venue of 'MA Chidambaram Stadium, Ch
venues = df[df["venue"]=="MA Chidambaram Stadium, Chepauk"]
wins = venues["winner"].value_counts()

sns.barplot(x=wins.index, y=wins.values)
plt.title('Number of Wins of Each Team at MA Chidambaram Stadium, Chepauk', fontsize=16)
plt.xlabel('Teams')
plt.ylabel('Wins')
plt.xticks(rotation=90)
plt
```

Out[54]:

```
<module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-packages\\matplotlib
\\pyplot.py'>
```



In [57]:

```
#8. Find and list the venues where the match resulted in a tie
ties = df[df["result"]=='tie']
venues = ties["venue"]

venues
```

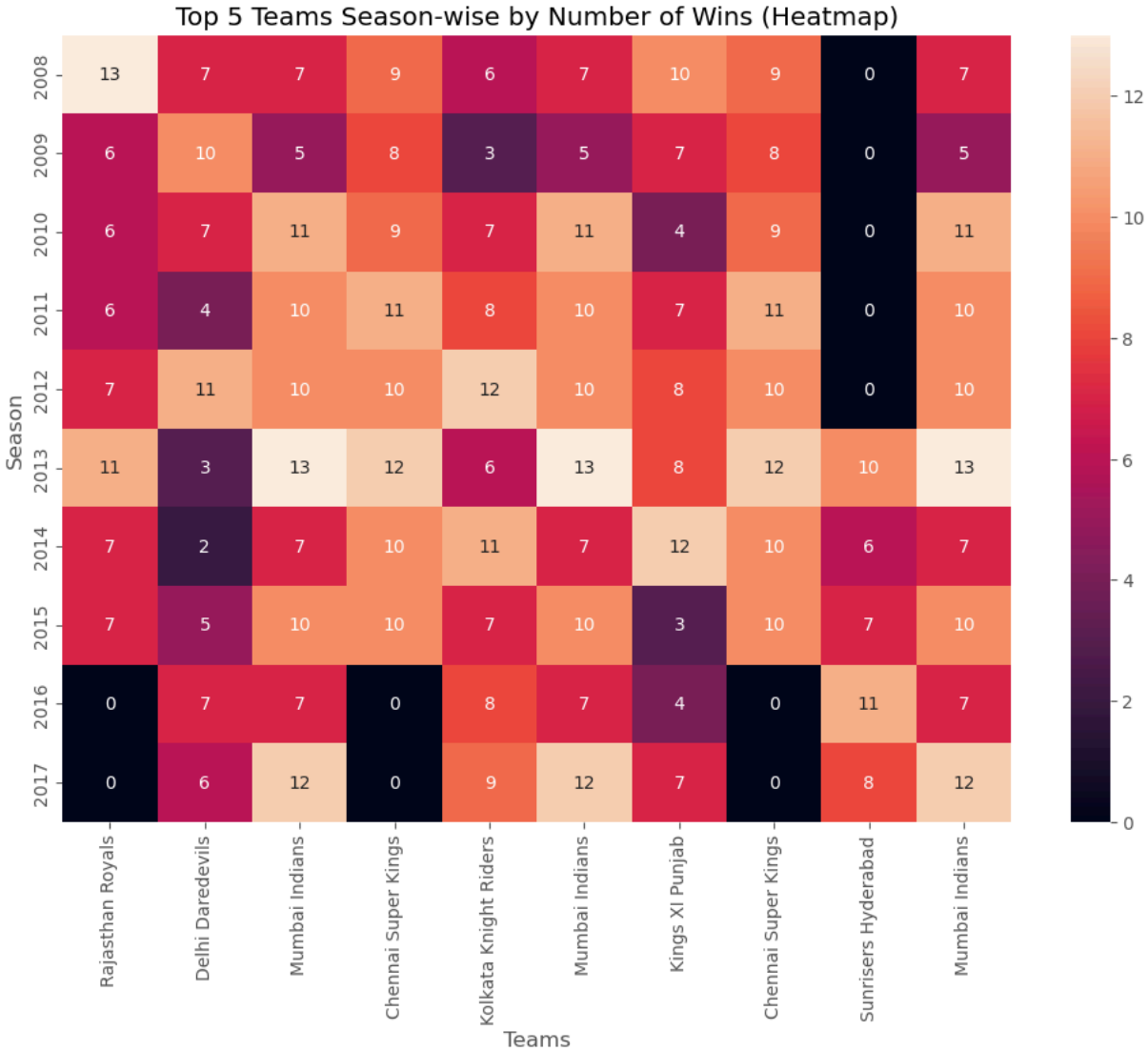
Out[57]:

```
33        Saurashtra Cricket Association Stadium
125                                   Newlands
189              MA Chidambaram Stadium, Chepauk
387    Rajiv Gandhi International Stadium, Uppal
400                        M Chinnaswamy Stadium
475                         Sheikh Zayed Stadium
535                 Sardar Patel Stadium, Motera
Name: venue, dtype: object
```

In [59]:

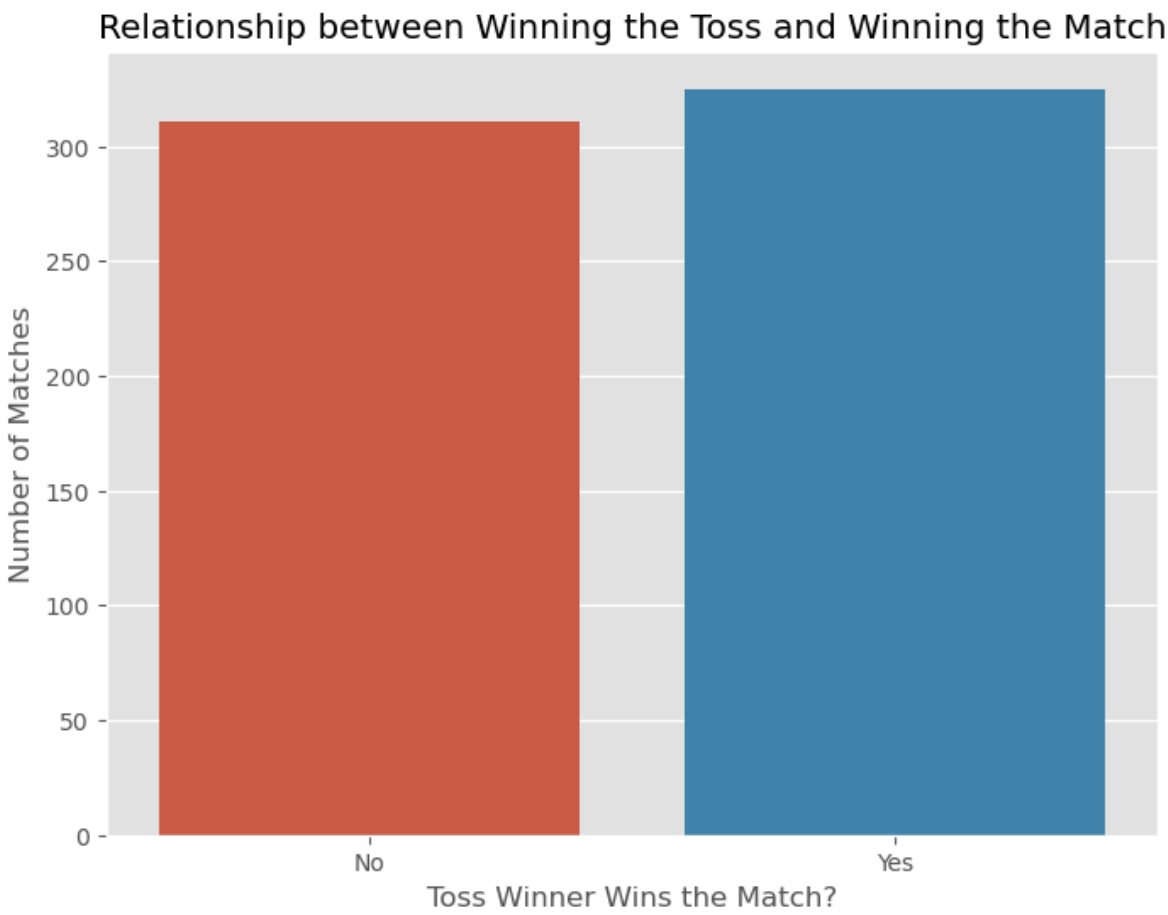
```
#9. List and visualize the best teams(top 5), season-wise
top = df.groupby(['season', 'winner']).size().reset_index(name='wins')
pivot_table = top_teams_season.pivot_table(index='season', columns='winner', values='wins'

top1 = pivot_table.apply(lambda x: x.nlargest(5).index.tolist(), axis=1)

plt.figure(figsize=(12, 8))
sns.heatmap(pivot_table[top_teams_season_sorted.apply(lambda x: x[0])], annot=True)
plt.title('Top 5 Teams Season-wise by Number of Wins (Heatmap)')
plt.xlabel('Teams')
plt.ylabel('Season')
plt.xticks(rotation=90)
plt
```

Out[59]: <module 'matplotlib.pyplot' from 'C:\\ProgramData\\Anaconda3\\lib\\site-packages\\matplotlib\\pyplot.py'>



In [61]:

```python
#10. Analyse and visualize the relationship between winning the toss and winning the match
df['toss_win_match_win'] = df['toss_winner'] == df['winner']

toss_win_match_win_count = df['toss_win_match_win'].value_counts()

plt.figure(figsize=(8, 6))
sns.barplot(x=toss_win_match_win_count.index, y=toss_win_match_win_count.values)
plt.title('Relationship between Winning the Toss and Winning the Match')
plt.xlabel('Toss Winner Wins the Match?')
plt.ylabel('Number of Matches')
plt.xticks([0, 1], ['No', 'Yes'])
plt.show()

toss_win_match_win_count
```



Out[61]:

```
True     325
False    311
Name: toss_win_match_win, dtype: int64
```

In [ ]:

```python

```